INVERSTIGATION REPORT

1. Introduction

Network traffic analysis is an important activity in network administration and cybersecurity. It involves examining packets transmitted across a network to understand communication between devices, troubleshoot connectivity problems, and identify potentially unusual network behavior.

This project uses Wireshark on Windows 11 to investigate DNS, ICMP, and TCP traffic. The investigation focuses on observing network protocols, interpreting packet information, and documenting findings using evidence from packet captures.

2. Problem Statement

Network connectivity problems can occur because of DNS resolution failures, network configuration errors, blocked traffic, or unsuccessful TCP connections. Without appropriate monitoring tools, identifying the source of these problems can be difficult.

This project demonstrates how Wireshark can be used to inspect network packets and support systematic troubleshooting through evidence-based analysis.

3. General Objective

To analyze network traffic using Wireshark to understand network communication and support network troubleshooting.

4. Specific Objectives

- To capture network traffic generated by authorized network tests.
- To identify and analyze DNS, ICMP, and TCP packets.
- To apply Wireshark display filters to examine specific network protocols.
- To document investigation findings and recommend appropriate troubleshooting actions.

5. Tools and Technologies

- **Wireshark:** Packet capture analysis and network protocol inspection.
- **Npcap:** Packet capture support on Windows.
- **Windows 11:** Operating system used for the investigation.
- **Command Prompt:** Used to run network diagnostic commands.
- **GitHub:** Used to store and document the project.

6. Investigation Environment

The investigation is conducted on a Windows 11 computer connected to a network that the investigator owns or is authorized to inspect.

The following commands are used to generate and test network traffic:

```
ipconfig
nslookup example.com
ping -n 4 1.1.1.1
```

The following Wireshark display filters are used:

Filter	Purpose
dns	Displays DNS traffic
icmp	Displays ICMP packets
tcp	Displays TCP packets
tcp.flags.syn == 1	Displays TCP packets with the SYN flag set
tcp.flags.reset == 1	Displays TCP packets with the RST flag set

7. Investigation Procedures

7.1 Network Configuration Inspection

The ipconfig command is used to inspect the computer's IP address, subnet mask, default gateway, and DNS configuration.

The information provides a baseline for understanding the network environment before packet analysis.

7.2 DNS Traffic Investigation

The command nslookup example.com is used to generate a DNS lookup.

Wireshark is filtered using dns to examine the query and any corresponding response.

The investigator records the requested domain, visible response information, and whether the lookup completed successfully.

7.3 ICMP Connectivity Investigation

The command ping -n 4 1.1.1.1 sends ICMP echo requests to a destination.

Wireshark is filtered using icmp to examine the requests and any replies.

The investigator compares the packets with the Command Prompt results to assess whether responses were received.

7.4 TCP Traffic Investigation

Normal, authorized network activity, such as opening a website, is generated while Wireshark is capturing traffic.

The investigator applies the tcp filter and examines TCP flags, source and destination addresses, and connection establishment where visible.

The analysis may identify SYN, SYN-ACK, and ACK packets associated with a TCP three-way handshake.

8. Findings and Observations

Important: The following sections must be completed using the actual packet capture. No packet counts or investigation results are assumed in this report.

8.1 DNS Investigation

- **Filter used:** dns
- **Observation:** Record the DNS query and response observed.
- **Interpretation:** Explain what the observed packets indicate about name resolution.
- **Evidence:** Insert screenshot filename.
- **Recommendation:** If a response is missing, verify the selected interface, repeat the lookup, and check the system's DNS configuration.

8.2 ICMP Investigation

- **Filter used:** icmp
- **Observation:** Record whether echo requests and replies were observed.
- **Interpretation:** Compare packet observations with the ping results.
- **Evidence:** Insert screenshot filename.
- **Recommendation:** If replies are missing, check connectivity and filtering rules. A missing ICMP response alone does not prove the destination is unavailable.

8.3 TCP Investigation

- **Filter used:** tcp
- **Observation:** Record the TCP flags and connection packets actually observed.
- **Interpretation:** Explain whether the captured packets show successful connection establishment or another observed behavior.
- **Evidence:** Insert screenshot filename.
- **Recommendation:** Investigate retransmissions, resets, or unsuccessful connections only if they appear in the capture and are relevant to the problem.

9. Limitations

The investigation is limited to the traffic captured on the selected network interface and during the selected time period. Some packets may not be visible because of encryption, DNS caching, capture timing, or network filtering.

The absence of a packet or response does not independently prove a security incident. Additional tests and evidence may be necessary before drawing conclusions.

10. Security and Ethical Considerations

Only traffic that the investigator owns or is authorized to inspect is captured.

Packet captures and screenshots must be reviewed before publication to prevent the exposure of credentials, private information, internal hostnames, or other sensitive data.

11. Recommendations

- Verify IP addressing, gateway, and DNS configuration when troubleshooting connectivity.
- Use protocol-specific display filters to narrow investigations.
- Compare packet observations with results from network diagnostic commands.
- Maintain records of observations, screenshots, and troubleshooting steps.
- Review packet captures for sensitive information before sharing them publicly.

12. Conclusion

This project provides practical experience in network packet capture, protocol analysis, and evidence-based troubleshooting using Wireshark. By examining DNS, ICMP, and TCP traffic, the investigator develops skills useful in network administration and cybersecurity.

The final conclusions must be updated after the practical tests are completed and the captured evidence has been analyzed.

13. References

- Wireshark Foundation. Official Wireshark website: https://www.wireshark.org/
- Wireshark Documentation: https://www.wireshark.org/docs/
- Microsoft Support. Windows networking and command-line diagnostic tools: https://support.microsoft.com/